

Pregame Pitcher Strikeout Projection - Resume Project Summary

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Primary project link: [GitHub repository](#).

Secondary long-form write-up: [Technical manuscript](#)

Goal

Estimate a starter’s strikeout rate before first pitch, project how many batters that starter will face, and convert the pair into expected strikeouts and $P(K \geq L)$ for common prop lines L —using only information available before first pitch:

$$k_{\text{rate}} \times \text{TBF} \rightarrow E[K] \rightarrow P(K \geq L)$$

Professional positioning

This project demonstrates a blended profile across:

- **AI/ML engineering:** end-to-end feature pipelines, freeze discipline, and reproducible deployment artifacts.
- **Applied statistics:** chronological validation, calibration diagnostics, and leakage-safe modeling controls.
- **Quant decisioning:** market-skill ranking, policy-floor governance, and risk-aware deployment gates.

Models and headline metrics

Component	Model	Role
Strikeout rate (live)	Weighted 3-model LightGBM blend	Active production scorer (<code>live_krate_ensemble.json</code>)
Batters faced (TBF)	Ridge (thin bullpen, 24 features)	Projected exposure
Counts / lines	Binomial / Poisson on projected TBF	Expected K and $P(K \geq L)$

Active deployment profile: blend 0.00 sparse72 / 0.60 sparse72_monotone / 0.40 final58, edge floor 0.12, ROI **0.4363**, PnL **1208.55**, Sharpe **0.4438**, Sortino **0.4277**, max drawdown **0.1905**, and market-skill deltas **+0.2069** (Brier) / **+0.1551** (LogLoss).

Full-stack evaluation (production)

Evaluation gate	Result
Estimator tuning	Keep baseline LightGBM defaults; Ridge α tuned and persisted
Calibration + probability quality	Production isotonic path; Brier 0.2090, LogLoss 0.6087, ECE 0.0639, MCE 0.1353

Evaluation gate	Result
Population policy	Metrics conditional on PA ≥ 9 (~3.5% of first pitchers excluded)
Exogenous-exit governance	Source-tagged anomaly overrides + training mask are shipped; current walk-forward A/B effect is neutral under low historical tag density

Current governance winners

High-signal results from current artifacts:

- Open-universe deduped sweep winner (ensemble_sweep_ranked_ensemble_full_aug21_deduped.csv):
- blend 0.05 sparse72 / 0.45 sparse72_monotone / 0.50 final58
- ROI 0.6612, Sharpe 0.9468, Sortino 0.6954, bets 35
- Brier/LogLoss skill vs market: +0.0825 / +0.0645
- Active deployment king (open_top3_transfer_manual_replay_aug21_deduped_top3_from_dedupedsweep.json):
- blend 0.00 sparse72 / 0.60 sparse72_monotone / 0.40 final58
- ROI 0.4363, PnL 1208.55, Sharpe 0.4438, Sortino 0.4277, max DD 0.1905
- Brier/LogLoss skill vs market: +0.2069 / +0.1551
- Duplicate-ticket diagnostics explicitly tracked: 123 duplicate groups, 246 duplicate tickets in manual-set audits.

Resume-ready takeaway

Leakage-safe Statcast -> rate x exposure stack with nested chronological selection, then scaled into a quant-governed production system with open-market replay, deduped fairness controls, transfer calibration, and live ensemble deployment. The stack shows positive market-skill deltas in current governance lanes, and runs under explicit CI/gate discipline without overstating unresolved live-edge evidence.

Portfolio claim: production-minded ML engineering under a strict information constraint, strengthened by statistical rigor and quant-style decision governance.

Copy/Paste resume bullets

- Built a leakage-safe MLB forecasting pipeline over pitch-level Statcast data (Polars + Python), enforcing chronological splits and pregame-only features, then wired it into a governed live decision system.
- Designed and deployed a two-stage prediction stack (k-rate x projected TBF) with isotonic-calibrated line probabilities and artifact-backed governance diagnostics in production.
- Productionized a governed decision engine with isotonic calibration, board-to-ledger parity locks, and execution freshness/coverage gates; hardened daily run reliability with explicit fail-fast controls.
- Led model-selection governance using open-universe counterfactual replay and deduped one-opportunity-one-bet evaluation; top transfer profile delivered **0.436 ROI**, **1208.55 PnL**, and positive market-skill deltas (Brier +0.2069, LogLoss +0.1551) in current artifact-backed replay lanes.

Interview framing (30 seconds)

I built a production-first sports forecasting system where the hard part was not just model accuracy, but controlling leakage, calibration drift, and execution quality. The final stack combines a frozen LightGBM ensemble with a projected exposure model, then governs decisions with chronological testing and explicit risk gates so we can operate daily without over-claiming edge.

Code first: [GitHub repository](#) · *Paper:* [manuscript PDF](#)